Soil Sampling Team Training and Qualifications. Supporting Document `IndigoCarbon_US-1_2023_0007`

March 31, 2025

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Context

Indigo has transitioned away from direct management of soil sampling by both internal staff and contracted individuals and now contracts with external organizations to conduct the soil sampling activities. However, sampling is still done according to Indigo's Standard Operating Procedure (SOP) and using Indigo's proprietary iPad application (Indigo Fields Mobile) for sample point generation, locating, and data collection.

This document includes information regarding both the soil sampling in the current reporting period as well as the historical information regarding past soil sampling processes and activities.

Current Sampling Team Training & Qualifications

For soil sampling conducted in 2024, Indigo contracted with Yard Stick PBC (https://www.useyardstick.com/). Key milestones with this vendor and other details related to training and qualifications are listed below:

- Indigo contracted with Yard Stick PBC in late 2023.
- The primary points of contact (POC) were Kevin Spence for Indigo (no longer with Indigo) and Max Slater for Yard Stick.
- On January 17, 2024 Kevin Spence e-mailed the Yard Stick team the key documents needed for soil sampling: SOPs for 30 cm sampling of carbon, bulk density, and pH and texture. These attachments are included as the following supporting documents:
  - `IndigoCarbon_US-1_2023_0001_v4.1_FINAL - Soil Carbon (30 cm) Sampling`
  - `IndigoCarbon_US-1_2023_0002_v4.0_FINAL - Bulk Density (30.48 cm) Sampling`
  - `IndigoCarbon_US-1_2023_0003_v4.0_FINAL - pH and Texture Composite (30 cm) Sampling`

- On February 15, 2024 the Indigo team conducted a virtual, 90-minute training, led by Adriel Hsu-Flanders (no longer with Indigo). The agenda for the training was the following:
  - 15 - 20 min demo on web app (Adriel)
  - Q&A on web app, and you all test on your own devices
  - 15 min demo on mobile app (Adriel)
  - Q&A and you all load up the app and test on your own devices
  - Post-training: we will assign you test tasks on mobile, which everyone will be asked to complete from home (estimated time commitment 20 min)
- Participants from Yard Stick included:
  - Kelsey Chan
  - Maya Lefelman
  - Charlotte Mondale
  - David Parris
  - Max Slater
  - Isaac Ubben
  - Seneca Wolman

Historical Information

Context

To ensure soil sampling was conducted in a standardized manner and executed appropriately, the following minimum qualifications were used to source soil samplers. In addition to the formal training outlined in the "Training Schedule," an Indigo representative was made available on an informal regular weekly basis during the main sampling seasons to answer sampler questions and clarify SOP requirements.

Sampler Qualifications:

The following general guidelines are used when selecting samplers for the Carbon by Indigo program:

Minimum qualifications:

- Two-year degree
- Must own/able to operate a vehicle that can pull 3500 lbs
- Agriculture experience
- Ability to travel seasonally, 80% during peak sampling

Core Competencies:

- Great communication skills and establishing relationships with growers
- Comfort with uncertainty with travel scheduling
- Ability to drive ATV from point to point to pull accurate soil tests
- Be comfortable and prepared for all-weather conditions, long/overnight hours of road travel and navigation
- Familiarity with operating and troubleshooting mechanics of augurs', soil probes and collection devices
- Able to lift >30 lbs

- Attention to detail and able to be familiar with protocols and following detailed directions
- Stringent about following safety protocols
- Use best judgment on schedules and workload
- Work with scheduling and geofencing software
- Work as part of a team and stick with timelines
- Self-driven and motivated with high ownership in the project to drive success
- Flexible and willing to adapt to changes in protocol
- Ability to adapt and learn new technologies and techniques with growth of a project

Formal Training Schedule

At the beginning of major field sampling seasons (e.g. spring prior to summer crop planting, fall post-summer crop harvest, etc.) formal training sessions were scheduled with samplers to review general sampling protocols, train on new SOP updates, and provide an opportunity for samplers to ask questions/receive clarification on the soil sampling program. The following outlines the dates and major topics covered of those meetings.

Table 1. Formal training schedule

Date	**Main Updates**	**Trainer**
October, 2019	Original Sampler Trainings	Lauren M., Dan Ochs, Brian Bartle, Dan Mongeau
December 4, 2019	• Soil samples should be shipped within 4 days of collection • Sampling under wet or frozen conditions	Derrick Anderson and Juan Cantu Supported by: Jay Weeks
March 18, 2020	• General protocol review • Bulk density sampling density increased from 1:5 to 1:3 (bulk density samples: carbon samples) • Bulk density sites selected using die instead of sequential count • Addition of texture/pH composite sample (1 per field)	Jay Weeks
April 29, 2020	• ERROR Protocol introduced to document reason why individual points cannot be sampled	Jay Weeks
May 12, 2020	• How to record tillage events at point-level when tillage is inconsistent • Simplified recommended sample labeling procedure introduced • Review of shipping requirements	Jay Weeks
September 16, 2020 (Clarified September 21, 2020)	• General protocol review • Sampling relative to tillage protocol added • Sampling points prescribed in IF instead of selecting bulk density points using die • Offsetting protocol revisited	Jay Weeks
Vendor Trainings February 25, 2021 – 2 vendors February 26, 2021 – 1 vendor March 31, 2021 – 1 vendor	• General protocol training • 2021 updates noted in appropriate supporting docs	Tobin Hoffman, Russell Booth, Jackie Boden

Date	Main Updates	Trainer
September 30, 2021	• General Protocol Overview • Covered protocol & IF additions with all vendors	Russell Booth
March 31, 2022	• Protocol review • IF enhancement training	Russell Booth
August, 2022	• IFM refresher training	Tiffany Harville

Informal Weekly Indigo Representative

During periods of high field sampling activity, informal group telemeetings generally occurred on a weekly cadence where samplers and/or sampler managers could receive feedback and ask questions. This often is helpful to confirm protocol questions and ensure sampling is being executed as uniformly as possible.

October 1, 2019 – February, 28, 2021: Jay Weeks

March 1, 2021 – April: Russell Booth

April 2022 – Present: Tiffany Harville

Summary of Changes

Section	Date	Summary of Changes
Current information	Dec 9, 2024	Section created and all content added